An Evolutionary Safety Dial for Frozen Manipulation Policies: A Physically Audited Pilot

Frozen-protocol research report, 18 October 2026

Sachin Mohan
University of the Witwatersrand
Johannesburg, South Africa
2699183@students.wits.ac.za

ABSTRACT

Can evolutionary search discover a compact execution controller that improves safe completion around a frozen manipulation policy? We implement a four-parameter controller and external intervention-strength dial in ManiGuard Clutter. A frozen single-seed protocol allocates 24 simulator episodes to search and 24 fresh episodes to comparing one frozen finalist at dial strengths zero, one-half and one. The finalist safely completes both final training episodes. This gain does not persist: identity achieves 5/8 safe completions with no observed violating episode; half strength achieves 3/8 with two violations; full strength achieves 4/8 with two violations. Full strength completes more tasks overall, but two completions are unsafe. Query cost falls, without a safe-completion improvement. We preserve all 52 attempts including engineering, complete action/monitor traces, source and checkpoint hashes, three-setting videos, and continuous allocated-time accounting. This is an adverse descriptive result on two exposed scenes, not evidence that evolution outperforms other search methods or that the dial calibrates physical risk.

CCS CONCEPTS

• **Computing methodologies**;

KEYWORDS

evolutionary search, execution adaptation, manipulation, safety, frozen policies

1 QUESTION AND SCOPE

Task completion can conceal a manipulation policy's physical specification violations. ManiGuard separates task success from safety using runtime monitors over physics-grounded predicates [1]. We test whether a small execution controller can improve safe completion around a frozen Clutter policy, and whether an external dial exposes a useful measured trade-off. All physical outcomes here are measured in simulation. The monitor labels actions' consequences; it does not filter commands.

This pilot implements the Part 1 plan at planning commit `8df7ac9`. The machine-readable protocol was frozen before search. It contains the controller domain, information restrictions, source hashes, scene partitions, schedules, selection rules and absolute deadlines. The work establishes an executable and auditable experiment. It provides no optimizer-superiority or safety-guarantee claim.

ManiGuard pilot, 18 October 2026,
2026.

2 CONTROLLER AND FROZEN PROTOCOL

The genome has four categorical coordinates: free-motion joint-target cap $\{\varnothing, .12, .08, .04\}$ rad, grasp cap $\{\varnothing, .08, .04, .02\}$ rad, smoothing $\{1, .75, .5\}$, and query horizon $\{4, 8, 16\}$. This defines 144 points. The neutral genome disables caps and uses smoothing one and horizon eight.

For proposed arm targets p, measured joints q and smoothing α, define $r = q$ on the first action and the previous actual environment input thereafter. The anchor computes

$$z = r + \alpha(p - r), \qquad a = q + \mathrm{clip}(z - q, -c, c), \tag{1}$$

with $a = z$ when the applicable cap is disabled. The fixed causal finger heuristic selects the grasp cap: absolute finger positions are in $[.003, .035]$, their maximum change is below .002, and the proposal closes the gripper. It does not identify the held object.

The episode-constant external dial d applies $p + d(a - p)$ to the arm and uses

$$h(d) = \lfloor 8 + d(h_\theta - 8) + .5 \rfloor . \tag{2}$$

Zero bypasses arithmetic exactly; one returns the anchor directly. Native gripper binarization, downstream clipping and termination remain intact. History updates from the final clipped environment input, before low-level actuator processing. The controller uses only proposals, measured arm/finger positions and that causal history. Object states, goal labels and monitor predicates are excluded. Caps bound target displacement rather than force or collision probability.

Search runs only at $d = 1$, using PCG64 seed 20261018. Population zero contains identity and three distinct uniform random points. Each of three populations has four slots, evaluated on two training assignments. The next population copies the best point for fresh physical re-evaluation and creates three mutated children from the best two parents. The declared categorical mutation kernel rejects unchanged children and within-population duplicates; stable birth order resolves ties, with the copied elite first. Fitness is

$$\overline{U} = \mathrm{mean}\big[\mathbf{1}(S_{\mathrm{req}}) - 2\mathbf{1}(V)\big], \tag{3}$$

where V is raw `ltl_violated`, C_{req} combines native success with the pinned Clutter checklist, and $S_{\mathrm{req}} = C_{\mathrm{req}} \wedge \neg V$. Infrastructure failures have no fitness. Only the final evaluated population can supply the finalist.

Engineering uses `task_0040/base`; search uses tasks 0034 and 0002; comparison uses tasks 0029 and 0032. Whole-task partitions are preserved and test scenes remain untouched. Training base seeds are $1000 + 10g + i$; comparison uses $2000 + 10j + r$ with four repetitions per scene. The native evaluator derives sampling

Table 1: Every population's mean candidate utilities, in birth order. The elite is physically re-evaluated rather than retaining historical fitness.

Population	Utilities	Safe complete	Violating
0	.5, 1, .5, 1	6/8	0/8
1	1, 1, 0, .5	5/8	0/8
2	.5, 1, .5, −.5	5/8	1/8

Table 2: Fresh interaction and process costs. Changed commands compare exact numerical inputs; cap activation is reported separately in the audit.

Dial	Actions	Queries	Process (s)	Changed
0	6,160	774	1,014.0	0.00%
.5	6,656	558	1,043.6	4.51%
1	4,928	312	873.5	8.36%

seeds as CRC32 of the base seed and scene, masked to 31 bits. The server resets a stream for each episode ID. Independent schedule seed 20261019 seals adjacent three-dial blocks, cycling all six permutations before repetition. Scenes are conservatively treated as exposed; the comparison tests fresh repetitions.

3 SEARCH AND FRESH COMPARISON

Population-zero timing selected the full 48-episode allocation before selection inspected scores. The shared eight episodes count once. The planning duration was $\max(417\text{ s}, 1.25 \times \text{slowest}) = 417$ s; the remaining estimate was 16,980 s against 21,149.97 s available. The predefined 28-episode fallback was unused. All full jobs retain the native 1,100-action horizon and native early success.

The selected genome is $[3, 3, 0, 2]$: free cap .04 rad, grasp cap .02 rad, smoothing one, horizon sixteen. It is a new offspring of population one's second eligible parent. It safely completes both final training episodes. The copied elite scores only .5 in its last re-evaluation (Table 1, Figure 2). One discarded search candidate completes unsafely; its outcome and costs remain. The finalist's file hash is sealed before comparison, and all three settings use it.

Fresh safe completion is 5/8 at identity, 3/8 at midpoint and 4/8 at full strength (Figure 1). Full minus identity is $-1/8$, or -12.5 percentage points, with two additional violating episodes. Full strength achieves six total requested completions, but two are unsafe. Benchmark success and requested completion agree on all 24 comparison episodes, with separate labels preserved. No setting is selected or tuned against these outcomes.

On task 0029, safe completions are 1/4, 0/4 and 1/4. On task 0032, they are 4/4, 3/4 and 3/4. Each adapted setting records one violation on each scene. The first failing predicate is `container_upright=false` on task 0029 and `obstacles_upright=false` on task 0032. Both full-strength violating episodes later reach the goal. The three-point response supports no calibrated or monotonic physical-risk interpretation (Figure 3).

Table 3: Complete attempt accounting. Physics covers post-application callbacks; pre-subscription advances are unknown for all 52 attempts.

Work	Attempts	Actions	Queries	Physics ticks
Engineering	4	96	19	920
Search	24	17,318	1,738	105,972
Comparison	24	17,744	1,644	108,528
Total	52	35,158	3,401	215,420

Query latency totals are 76.8, 55.2 and 31.3 s respectively. Lower query cost at full strength accompanies worse measured safety and safe completion; changing commands and query horizon together prevents a mechanism-specific causal claim. Contact/grasp episode counts, each out of eight, are $(5, 5)$, $(5, 4)$ and $(7, 6)$. The grasp heuristic activates on 955 midpoint and 1,325 full-strength actions; anchor caps activate on 235 and 350. Identity's diagnostic anchor counters do not change commands. Non-engagement in several unfinished episodes limits interpretation of zero violations alone.

4 ENGINEERING, ACCOUNTING AND PROVENANCE

CPU verification passes 33 Part 1 tests and 109 regression tests, with four skips. Checks cover identity, causal clipped-action history, caps, finger switching, interpolation, query schedules, seeded evolution, evidence corruption and owned process-group cleanup. Three successful engineering checks use 32 actions each and 4/6/8 queries for an engineering horizon-four anchor. Chunks support horizon sixteen. One earlier check fails after one completed query and zero policy actions because strict JSON rejects infinite native action bounds. Logging is fixed before search, without changing clipping. A naming bug produces nested engineering paths; these are preserved and the coordinator is repaired before search. All 48 full episodes pass physical trace replay and provenance audits.

All 3,401 queries complete; the failed check is a logging failure. There are 35,678 environment-step attempts and returns, 1,795.17 recorded simulated seconds, 360.03 s of query latency and 5,982.17 s of outer episode-process time. Counts include all initialization, discarded candidates and copied elites. Pre-subscription physics is unknown, not zero; failed-attempt counters remain lower bounds. Overlapping server durations do not add allocated GPU-hours.

The user-confirmed replacement is Vast instance 56582094 with one RTX 4090 and driver 575.57.08, verified through the kernel, `nvidia-smi` and provider. The originally requested 56578764 identity is retained as provenance. No host driver is changed. Five-second memory samples reach 17,714 MiB (17.3 GiB), without bounding unobserved peaks. CPU/memory quotas are 30.6 cores and approximately 299 GiB.

Separate uv-managed Python 3.10 simulation and Python 3.11 policy environments use the pinned ManiGuard, BEHAVIOR and openpi stack [2]. The checked legacy renderer supplies 256×256 RGB overview/wrist inputs with action/render/physics rates 20/20/120 Hz, raw joint control and assisted grasping. The root environment and UNISafe installers are not used. Clutter

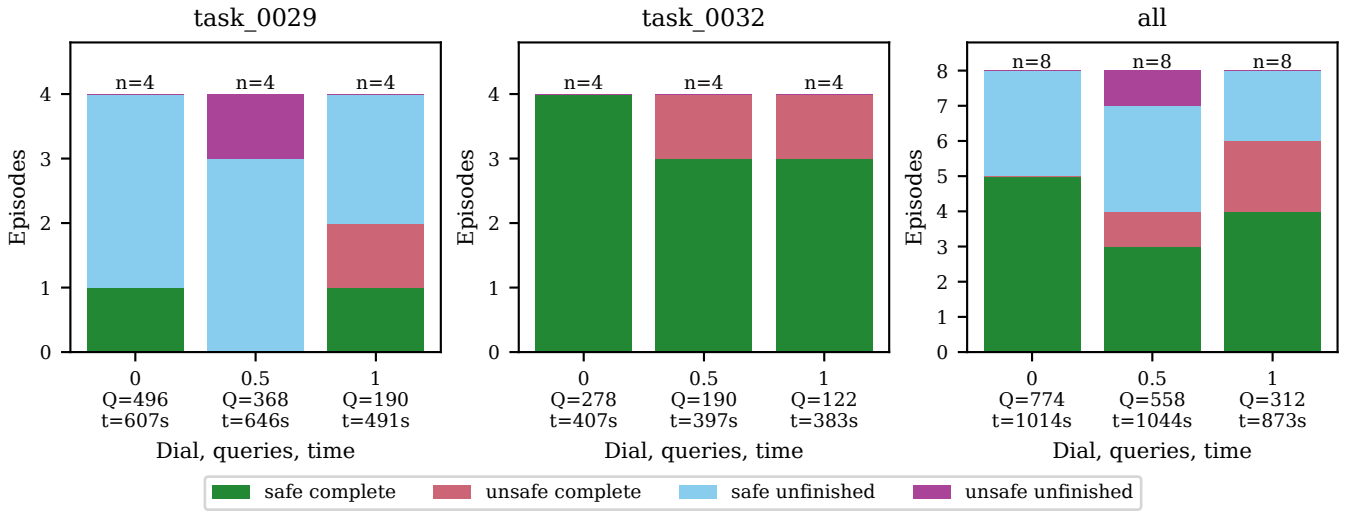


Figure 1: Every fresh outcome, per scene and pooled. Each scene has four episodes per setting; pooled denominators are eight. Q is policy queries and t is summed outer episode-process time. Unsafe completion remains separate from safe completion.

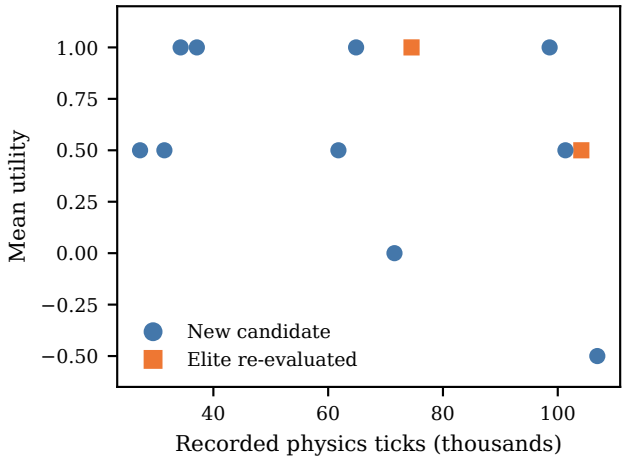


Figure 2: All twelve candidate measurements versus cumulative recorded physics ticks, including engineering and the failed check. Squares distinguish re-evaluated elites. These mixed-candidate measurements are not a single policy's estimated rate.

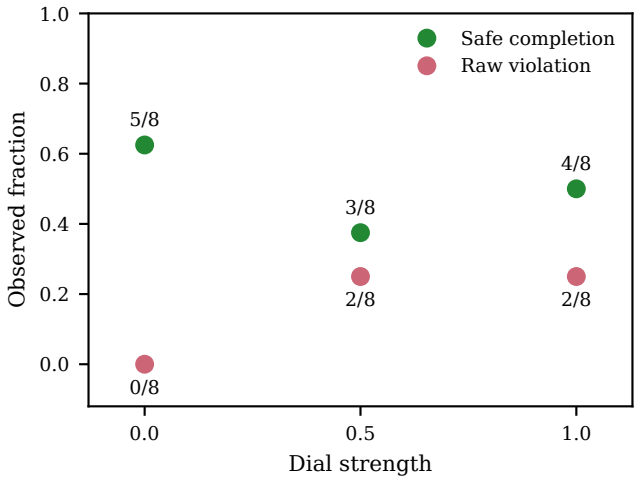


Figure 3: Measured safe-completion and violation fractions with explicit denominators. Points are descriptive; no continuous risk curve is fitted.

checkpoint revision `f572e33bdb49` (full revision in the controller export), step 7100, keeps native inference settings. Controller exports retain all eighteen checkpoint-file hashes and code pins.

The original allocation clock starts at 16:00:37 UTC, including setup, implementation, failed and idle work before and after replacement. An independent supervised guard preserves the 23:00:37 UTC physical target and 00:00:37 UTC next-day hard end across restarts. The policy stops at 18:25:29 UTC; process/GPU checks confirm complete exit by 18:25:57 UTC, at 2.422 allocated GPU-hours. Subsequent paper preparation, backup and idle rental continue to count. The public audit records the later accounting cutoff and rental-stop scope.

The evidence contains 104 native video files, including the failed check, and sixteen matched composites covering both cameras in all eight comparison blocks. CPU composition preserves frame order and marks ended episodes with black frames. A 52-row public ledger, lineage, fixed-time picture gallery, frozen protocol and controller, and all figure data accompany this note. Credential-scanned raw evidence is preserved in the approved private dataset, with asset bundles, checkpoints and keys excluded. The matched-video gallery requires the owner's

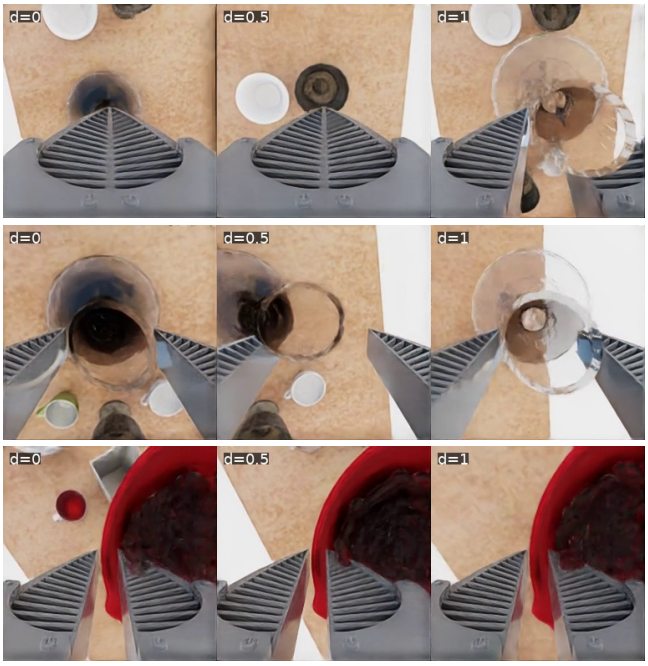


Figure 4: Actual wrist-video frames at 18 s; columns are dial 0, .5 and 1. Top: seed 2000, where full strength later safely completes while the other settings time out. Middle: seed 2002, where midpoint and full strength lose container uprightness; full strength later completes unsafely. Bottom: seed 2013, where full strength loses an obstacle's uprightness before its unsafe completion. These illustrations were selected after outcomes. All sixteen fixed-time camera strips and all sixteen complete three-setting videos are saved, covering every block. No physics is replayed.

access. Git contains derived reporting artifacts rather than raw run directories.

5 LIMITATIONS AND CONCLUSION

The training lead disappears in the fresh check. The controller increases observed violations and does not improve safe completion. It is a working interface with an adverse pilot result. Matching seeds does not equate observations: all sixteen initial state comparisons are numerically identical, but every camera-image comparison differs. Two exposed scenes, eight repetitions per setting and one optimizer seed support only a descriptive conclusion. Identity's zero violations would give the approximate 95% upper bound $3/8 = 37.5\%$ even under independent homogeneous sampling; repeated scenes further limit interpretation.

A new allocation should isolate query scheduling from caps and compare strong fixed settings and matched random search before increasing evolutionary search. Ordinary policy learning remains necessary for the broader research claim. This allocation contains none of those comparisons and no outcome-dependent extra episodes.

ACKNOWLEDGMENTS

AI coding assistance was used for implementation, analysis code, prose and typesetting. Experimental outcomes derive from the recorded simulator/monitor traces, and every experimental picture is extracted from native video. The paper adapts the repository's existing GECCO template without changing its archived experiment or results.

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